



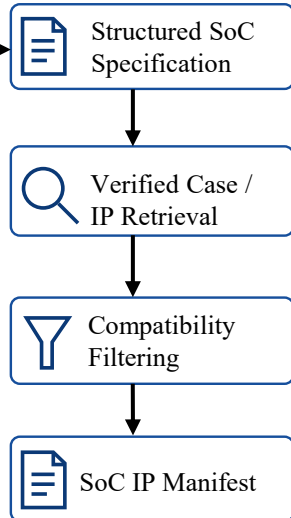
LLM-Assisted Orchestration Layer

Reasoning · HLS code generation · Design decisions · Tool invocation · Diagnostic-driven repair

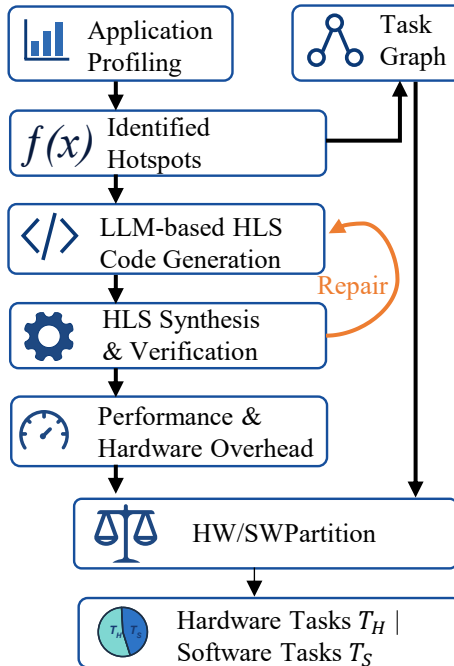
1 Requirement Interpretation & IP Selection

Natural-language Requirements

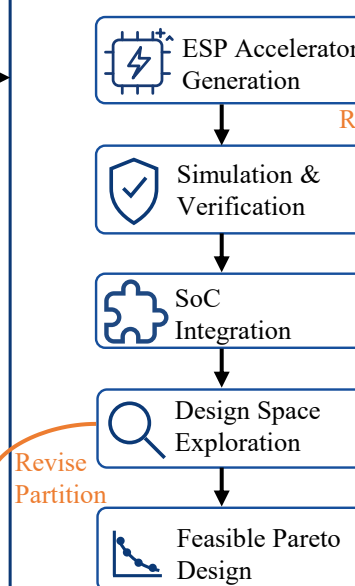
C/C++ Application



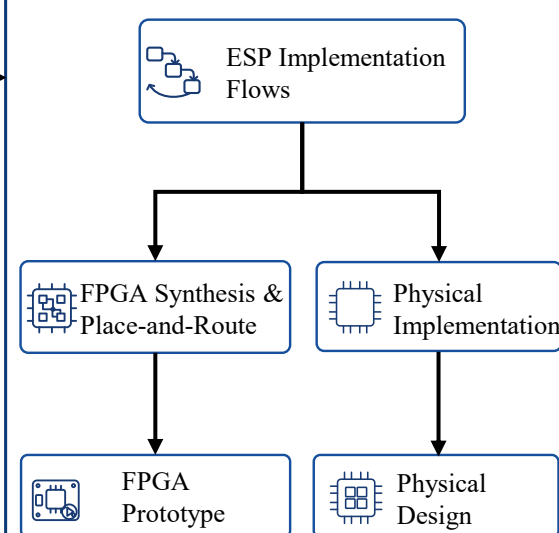
2 Profiling & HW/SW Partitioning



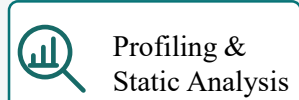
3 SoC Generation & System-level DSE



4 FPGA Prototyping & Physical Design



Quantitative Evidence and Tool Feedback



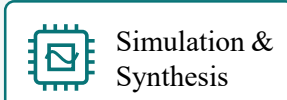
Profiling & Static Analysis



CPU Simulation



HLS Toolchain



Simulation & Synthesis



Design Space Exploration



ESP Toolchain

→ Primary Design Flow

- - -> LLM Control and Tool Feedback

→ Iterative Repair or Repartitioning